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Outdoor Activity: User Mental Model & Behaviour Study

Introduction:

The purpose of this activity was to observe how users interact with machines and equipment in a university environment and identify differences between what users expect and what actually happens during interaction.

Two machines were selected for the study:

- Smart Vending Machine
- Interactive Touchscreen Machine

The observations were supported by geotagged photographs taken at the university.

Machine 1: Smart Vending Machine

Machine Description:

The smart vending machine allows users to select products through a touchscreen, make digital payments, and collect the purchased item.

User Expectation:

The user expects the machine to provide a simple and straightforward process for selecting, paying for, and collecting a product. The interface is expected to clearly communicate each step.

User Behaviour:

- The user visually examined the products available inside the machine.
- The user looked at the touchscreen to identify and select the required product.
- The user referred to the instructions displayed on the machine.
- The user had to shift attention between the physical products and the touchscreen.
- The user examined the machine to understand how the purchased product would be collected.

Mental Model Mismatch:

The user expects product selection, payment, and dispensing to follow a direct and familiar vending-machine process. However, the product selection and payment take place through the touchscreen, while collection involves a separate compartment. This can create uncertainty for a first-time user.

Problem:

The product collection process is not sufficiently clear from the physical layout of the machine. The different interaction points can make the overall process less immediately understandable.

Improvement:

1. Provide a clearly visible step-by-step interaction flow covering product selection, payment, and collection.
2. Display clear feedback after payment indicating where and how the user should collect the product.

Evidence:



Photo 1: User interacting with the vending machine and touchscreen.



Photo 2: Front view showing the products, touchscreen, instructions, and collection area.

Machine 2: Interactive Touchscreen Machine

Machine Description:

The second selected equipment is a large touchscreen-based machine designed for providing digital information through an interactive display.

User Expectation:

The user expects the machine to clearly communicate its purpose and provide an obvious starting point for interaction. The available options and actions should be easy to identify.

User Behaviour:

- The user approached the touchscreen.
- The user examined the display to understand its purpose.
- The user looked at the screen before beginning the interaction.
- The user had to determine how to proceed from the available interface.
- The large touchscreen created an expectation of direct interaction.

Mental Model Mismatch:

The user expects a touchscreen-based machine to immediately provide clear interactive options. However, the initial interface does not make the available actions sufficiently obvious, creating uncertainty about how to begin.

Problem:

The main usability issue is poor discoverability. The user may not immediately understand the purpose of the machine or the first action required.

Improvement:

- Provide a clear initial interface that communicates the purpose of the machine.
- Use prominent navigation controls and interaction cues to make the available actions easier to identify.

Evidence:



Photo 3: User standing beside the interactive touchscreen machine.

Comparative Analysis:

Aspect	Smart Vending Machine	Interactive Touchscreen Machine
Purpose	Automated product purchase	Digital information access
User Expectation	Easy product selection, payment and collection	Clear starting point and easy interaction
User Behaviour	Examined products, touchscreen and instructions	Examined the touchscreen to understand how to proceed
Mental Model Mismatch	Collection process differs from the expected vending-machine interaction	Starting point and available actions are not immediately clear
Usability Problem	Unclear product collection process	Poor discoverability
Improvement	Clear interaction flow and collection feedback	Clear initial interface and interaction cues

Key Findings:

- Users rely on their previous experiences when interacting with machines.
- Differences between expected and actual behaviour can cause hesitation and uncertainty.
- Physical and digital interaction elements should provide a consistent flow.
- Clear instructions and feedback can reduce user confusion.
- The first interaction should be immediately understandable.

Conclusion:

The study demonstrates the importance of designing machines according to the user's mental model. The smart vending machine presents a mismatch in the product collection process, while the interactive touchscreen machine presents a mismatch in discoverability and initial interaction.

Both machines can provide a more intuitive user experience through clear interaction flows, visible instructions, effective feedback, and easily identifiable controls.